

Problem 4

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Problem 1 A positive continuous function will have area approximated on the interval $[1, 6]$ using n rectangles.

- (a) Find a formula for the grid point, x_k .

Explanation. Here $\Delta x = \frac{b-a}{n} = \frac{5}{n}$.

The basic grid point formula is $x_k = a + k\Delta x = 1 + \frac{5k}{n}$.

- (b) Find a formula for the sample point x_k^* if using a right Riemann sum.

Explanation. For right endpoints, $x_k^* = x_k = 1 + \frac{5k}{n}$.

- (c) Find a formula for the sample point x_k^* if using a left Riemann sum.

Explanation. For left endpoints, $x_k^* = x_{k-1} = 1 + \frac{5(k-1)}{n}$.

- (d) Find a formula for the sample point x_k^* if using a midpoint Riemann sum.

Explanation. For midpoints, $x_k^* = \frac{x_k + x_{k-1}}{2} = 1 + \frac{5(k - \frac{1}{2})}{n}$.